

# MOHAB AHMED

Dar misr, Al Obour, Cairo, Egypt

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## Education

**Egyptian Chinese University**

*Bachelor of Software engineering*

**Sep. 2022 – Expected june 2027**

*Cairo, Egypt*

## Relevant Coursework

- Data Structures
- Object-Oriented Programming
- Algorithms Analysis
- Database Management Systems
- Operating systems
- Software Engineering
- Software Testing
- Computer Architecture

## Academic Projects

### Tower Of Hanoi

**2023 – 2024**

- Developed a complete simulation of the Tower of Hanoi puzzle using Object-Oriented Programming and Data Structures principles in C++.
- Built a recursive algorithm that efficiently solves the puzzle for any number of disks and rods.
- Integrated input validation to handle edge cases such as zero or negative disk entries.
- Enhanced code modularity and memory management through header and implementation files (.h, .cpp, main.cpp). Displayed step-by-step disk movements and total move count to solve the puzzle.
- Given the most efficient project for solving the puzzle.

### TicTacToe

**2024**

- Designed and implemented a two-player Tic Tac Toe game using C with an intuitive user interface (Win Forms) / interactive CLI (Console).
- Developed core game logic including player turns, move validation, win/draw detection, and board state tracking.
- Utilized C features such as arrays, loops, conditionals, and event-driven programming for seamless gameplay experience.
- Implemented responsive UI (for WinForms) or clean and readable console output (for CLI) to ensure clear player interaction.
- Added modular structure with separate methods for game loop, move handling, win-checking logic, and board display.
- Ensured robustness with error handling for invalid inputs and continuous play options.

### Nutrition Management System | Python, Tkinter, Oracle SQL

**2024–2025**

- Collaborated with a team to Design and develop a full-featured nutrition management system using Python for backend logic, Tkinter for GUI development, and Oracle MySQL for database integration.
- Built user authentication module with secure login, registration, and password validation
- Implemented dynamic BMI calculator and automatic personalized diet suggestions based on user metrics (age, weight, height, activity level).
- nabled CRUD operations (Create, Read, Update, Delete) for meals, nutrition data, and user profiles with seamless MySQL connectivity via mysql-connector-python.
- Displayed nutritional breakdown, daily calorie tracking, and visual meal summaries using structured and user-friendly GUI components.
- Followed object-oriented design principles with modular, maintainable codebase (separate classes for User, Meal, NutritionLogic, and DatabaseManager).
- nabled CRUD operations (Create, Read, Update, Delete) for meals, nutrition data, and user profiles with seamless MySQL connectivity via mysql-connector-python.

### Sorting Algorithms Visualizer | python

**2025**

- Solo Project – Data Structures Algorithm Implementation, Developed a Python application to implement and demonstrate classic sorting algorithms: Insertion Sort, Merge Sort, and Quick Sort.
- Designed the system to output detailed step-by-step tracing of each algorithm's execution, enabling better understanding of: Key comparisons,Element swaps,Recursive calls and merging phases
- Each algorithm was implemented using a modular approach with separate functions, adhering to clean coding practices and time complexity principles.
- Compared algorithm performance on varied input sizes (best, average, and worst-case scenarios), highlighting time and space efficiency.
- Documented code with clear comments and docstrings to support educational use or integration into larger systems.

- collaborated with a team to design and develop a full-featured car service management system for booking, servicing, and client management across multiple branch locations.
- Built the frontend using Python with Tkinter, and integrated Oracle MySQL for backend data storage and operations using mysql.connector.
- Created and normalized a comprehensive relational database (3NF) with entities including Client, Garage, Branch, Car, Service, and Booking, ensuring data integrity through foreign key constraints.
- Implemented a dynamic booking system that links clients, cars, garages, and services via an optimized schema including intermediate tables (e.g., *Desired\_service*) to handle many-to-many relationships. Developed and executed complex SQL queries for CRUD operations, service history tracking, garage capacity management.
- Addressed real-world data design challenges: Ensured referential integrity across multiple interrelated entities, Normalized schema to reduce redundancy and improve efficiency, Designed scalable models to accommodate new services and branches.
- Provided intuitive GUI forms for client registration, service bookings, car entries, and real-time lookup of services by location.

## Technical Skills

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**Programming Languages:** Python, c, C++, SQL(Oracle, mySQL)

**Developer Tools:** VS Code, Anaconda, Pycharm, Visual studio

**computer skills:** Microsoft word, Microsoft Powerpoint, Chatbots, AI tools, vibe coder

## Soft Skills

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Team Work

Communication Skills

Time Management

Good learner

Patient

Problem Solver

Presentation Skills

## Languages

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Arabic (Native)

English (FLuent)

## Leadership / Extracurricular

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**Tickets Marchee — Event Organizing Team**

**Sep 2022 – dec 2023**

*Event organizer*

- Worked with Tickets Marchee to organize large-scale public events including concerts, festivals, and conferences.
- Handled logistics, ticketing, and on-ground operations to ensure successful event execution and customer satisfaction